
Making It Explicit

**Reasoning, Representing,
and Discursive Commitment**

Robert B. Brandom

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To Wilfrid Sellars and Richard Rorty

without whom most of it would not even be *implicit*

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. . . both a new world,
and the old made explicit . . .

We shall not cease from exploration
And the end of all our exploring
Will be to arrive where we started
And know the place for the first time.

T. S. ELIOT, "Four Quartets"

Contents

Preface

xi

PART ONE

1. Toward a Normative Pragmatics	3
I. Introduction	3
II. From Intentional State to Normative Status	7
III. From Norms Explicit in Rules to Norms Implicit in Practices	18
IV. From Normative Status to Normative Attitude	30
V. From Assessment to the Social Institution of Norms	46
VI. From Intentional Interpretation to Original Intentionality	55
<i>Appendix: Wittgenstein's Use of <i>Regel</i></i>	64
2. Toward an Inferential Semantics	67
I. Content and Representation	67
II. The Priority of the Propositional	79
III. Conceptual Classification and Inference	85
IV. Material Inference, Conceptual Content, and Expression	94
V. Circumstances and Consequences of Application	116
VI. Conclusion	132

3. Linguistic Practice and Discursive Commitment	141
I. Intentional States and Linguistic Practices	141
II. Deontic Status and Deontic Attitudes	157
III. Asserting and Inferring	167
IV. Scorekeeping: Pragmatic Significance and Semantic Content	180
4. Perception and Action: The Conferral of Empirical and Practical Conceptual Content	199
I. Assertions as Knowledge Claims	199
II. Reliability	206
III. Observation Reports and Noninferential Authority	213
IV. Rational Agency	229
V. Practical Reasoning: Inferences from Doxastic to Practical Commitments	243
VI. Intentions	253
PART TWO	
5. The Expressive Role of Traditional Semantic Vocabulary: 'True' and 'Refers'	275
I. From Inference to Truth, Reference, and Representation	275
II. Truth in Classical Pragmatism	285
III. From Pragmatism to Prosentences	299
IV. Reference and Anaphorically Indirect Descriptions	305
V. The Function of Traditional Semantic Vocabulary Is Expressive, Not Explanatory	322
6. Substitution: What Are Singular Terms, and Why Are There Any?	334
I. Multivalued Logic and Material Inference	334
II. Substitution, Sentential Embedding, and Semantic Roles	346
III. Subsentential Expressions	360
IV. What Are Singular Terms?	367
V. Why Are There Singular Terms?	376
VI. Objections and Replies	384
VII. Conclusion	399
<i>Appendix I: From Substitutional Derivation of Categories to Functional Derivation of Categories</i>	404
<i>Appendix II: Sentence Use Conferring the Status of Singular Terms on Subsentential Expressions—An Application</i>	409
7. Anaphora: The Structure of Token Repeatables	413
I. Frege's <i>Grundlagen</i> Account of Picking Out Objects	413
II. Definite Descriptions and Existential Commitments	432
III. Substitution, Token Recurrence, and Anaphora	449
IV. Deixis and Anaphora	459

V. Interpersonal Anaphora and Communication	473
<i>Appendix: Other Kinds of Anaphora—Paychecks, Donkeys, and Quantificational Antecedents</i>	490
8. Ascribing Propositional Attitudes: The Social Route from Reasoning to Representing	495
I. Representation and <i>De Re</i> Ascription of Propositionally Contentful Commitments	495
II. Interpretation, Communication, and <i>De Re</i> Ascriptions	508
III. <i>De Re</i> Ascriptions and the Intentional Explanation of Action	520
IV. From Implicit Attribution to Explicit Ascription	529
V. Epistemically Strong <i>De Re</i> Attitudes: Indexicals, Quasi-Indexicals, and Proper Names	547
VI. The Social-Perspectival Character of Conceptual Contents and the Objectivity of Conceptual Norms	584
<i>Appendix: The Construction and Recursive Interpretation of Iterated Ascriptions That Mix De Dicto and De Re Content Specifications</i>	608
9. Conclusion	614
I. Two Concepts of Concepts	614
II. Norms and Practices	623
III. We Have Met the Norms, and They Are Ours	639
<i>Abbreviations</i>	653
<i>Notes</i>	655
<i>Index</i>	717

Substitution: What Are Singular Terms, and Why Are There Any?

I start out from judgments and their contents, and not from concepts . . . I only allow the formation of concepts to proceed from judgments. If, that is, you imagine the 2 in the content of judgment $2^4 = 16$ to be replaceable by something else, by $\{-2\}$ or by 3 say, which may be indicated by putting an x in place of the 2: $x^4 = 16$, the content of possible judgment is thus split into a constant and a variable part. The former, regarded in its own right but holding a place open for the latter, gives the concept '4th root of 16' or 'the individual 2 falls under the concept "4th root of 16" or "belongs to the class of 4th roots of 16"'. But we may also just as well say '4 is a logarithm of 16 to the base 2'. Here 4 is being treated as replaceable and so we get the concept 'logarithm of 16 to the base 2': $2^x = 16$. . .

And so, instead of putting a judgment together out of an individual as subject and an already previously formed concept as predicate, we do the opposite and arrive at a concept by splitting up the content of a possible judgment.

FREGE, "Boole's Logical Calculus and the *Begriffsschrift*"

I. MULTIVALUED LOGIC AND MATERIAL INFERENCE

1. *Three Challenges for Inferential Approaches to Semantics*

The theoretical structure being explored here is animated by commitments both to a *deontic pragmatics* and to an *inferential semantics*. The first means that the states to be investigated, the original bearers of intentional contents, are to be understood normatively—more particularly as species of *commitments* and *entitlements*. The phenomenalist account of deontic statuses such as commitment, in terms of scorekeeping with the socially complementary deontic attitudes of *attributing* and *undertaking*, is offered as a way to begin filling in such an approach to pragmatics. The second theoretical commitment means that the *contents* that determine, in context, the deontic *significance* of adopting or altering a deontic status, or of performing a contentful act, are to be understood as broadly *inferential* roles. The content must, in context, fix the circumstances in which one would be entitled to adopt or undertake a commitment with that content and must fix the appropriate consequences of undertaking such a commitment. Employing an expression with that content then involves endorsing

the inferential commitment from those circumstances of entitlement to those consequences of commitment. The description of the game of asserting, of the inferentially articulated practices that confer assertible, that is, propositional, contents on states, acts, and utterances in virtue of their roles in that game, and the account of logical vocabulary as distinguished by its expressive task of making explicit as assertible contents precisely the inferential commitments that determine those roles, are offered as a way to begin filling in such an approach to semantics. This chapter continues the inquiry into inferential notions of semantic content.

There are three topics that would seem to pose special explanatory difficulties for attempts to understand semantic content in terms of proprieties of inference. First, the functional involvements that could plausibly be taken to be responsible for the conferral of such contents relate conceptually contentful deontic states not only to each other but also to the nondiscursive environment. Perception and action, as entries to and exits from the discursive realm, are governed by practical proprieties every bit as important as, and irreducible to, those governing purely inferential moves within that realm. So significant have the entries and exits seemed that each has been taken, by some empiricists and by some pragmatists in turn, to be the sole source of content for intentional states—to the exclusion not only of inferential articulation but of each other. How can a broadly *inferential* approach incorporate the aspects of semantic content conferred by the *noninferential* aspects of such entries and exits?

Second, the notion of content as inferential role seems naturally adapted to account only for *propositional* content, for it is only commitments with contents of this category that can play the role of premise and conclusion in inferences. But the sentences that express propositions typically have significant parts that are not sentences, which do not express propositions, and so which cannot serve as inferential premises and conclusions. Yet these subsentential expressions certainly ought to be said to be contentful, in virtue of what Dummett calls the “contribution” they make to the propositional contents expressed by sentences in which they occur. How can a broadly inferential approach to semantic content be extended from the grammatical category of sentences, the only sort of expression directly involved in inference, to various subsentential categories such as singular terms and predicates? For in the absence of contents corresponding to these categories, it would not be possible to understand important sorts of inferences, paradigmatically those codified explicitly by the use of identity and quantificational logical locutions.

Third, when the semantic theorist seeks to express conceptual contents explicitly, and so to reason about them—for instance when a question has arisen concerning how a certain remark should be interpreted—the semantic vocabulary employed includes, not only the logical locutions that have been construed as making *inferential* relations explicit, but also *representational*

locutions that should be understood as making referential relations explicit. Such locutions make it possible to say what someone is talking about, what is being referred to, what a belief is of or about, or what would make it true. How can a broadly inferential approach to semantic content account for the representational features of content that are expressed explicitly by means of such locutions?

The first of these *prima facie* difficulties has already been addressed (see Chapter 4). Although entry and exit moves are not themselves inferential moves, neither the noninferential acknowledgments of doxastic commitments that proximally terminate perceptual entries (as distinct from mere differential responses) nor the acknowledgments of practical commitments that noninferentially initiate actions (as distinct from other performances) can be understood apart from the role they play in the game of giving and asking for reasons, most directly as premises for cognitive reasonings, and as conclusions of practical reasonings, respectively. Perceptual reports are to be distinguished from mere reliable differential responses generally by their liability to demands for justification and their utility in providing justifications for other claims. Actions are to be distinguished from behavioral performances generally by their responsibility to assessment and deliberation concerning the inferentially articulated responsibilities they incur and discharge. So not only do perceiving that a content is true and acting so as to make it true involve endorsement of the inferential propriety of the move from the circumstances in which one is entitled to produce such a performance to the consequences one becomes committed to thereby, but those circumstances (of action) or consequences (of perception) themselves are inferentially significant.

The third of the cited challenges to an inferential approach to semantic content concerns its explanatory adequacy to the phenomena that make representational approaches to semantic content attractive and unavoidable. This is the most important and difficult issue. The general strategy for responding to it that is pursued here is to attempt to explain, in terms of the inferentially articulated social scorekeeping practices that institute discursive deontic statuses, what is expressed by the central sorts of representational semantic locutions. Where this can be done, the result is an account of what the theorist is *saying* when making claims about what represents what. Chapter 5 began this discussion by explaining the use of 'true' and 'refers' or 'denotes' (and so one crucial sense of 'represents') in terms of *anaphoric* links between expression tokenings. Chapter 8 completes the official treatment of representational locutions by specifying in discursive scorekeeping terms what it is to use locutions to make propositional-attitude ascriptions *de re*. This is the trope that makes it possible to specify what we are talking or thinking *of* or *about*, what objects our beliefs are directed at. This is the essential use in virtue of which expressions are properly interpreted as expressing attributions of ofness or aboutness in the intentional or semantic sense.

Both the discussion of what it is for a belief or claim to be of or about an object, or to be true of an object, and the discussion of 'refers', however, require that the inferentialist account of conceptual content be extended to subsentential expressions, paradigmatically singular terms and predicates. So the treatment of what is expressed by the central, explicitly representational locutions requires that the second challenge to the inferentialist order of semantic explanation be addressed. This should come as no surprise. For although some semantic thinkers (Davidson and Stalnaker are recent examples) conceive representational relations as obtaining in the first place between propositionally contentful intentional states and facts or states of affairs, they are in a distinct minority. Most representationalists have not taken the pragmatic priority of the propositional to entail a corresponding priority in the semantic order of explanation of conceptual contents. The more common position holds that the notion of representation is to be understood, to begin with, in terms of the representation of *objects*, particular things, and their properties and relations. According to this way of thinking, the basic representational bonds—in terms of which, for instance, the capacity for propositional representation, the capacity to represent possible states of affairs, is to be accounted for—are taken to be those linking represented objects to object-representings and represented properties to property-representings. If something like this turned out to be correct, adequate explanations of the function of attributions of representational purport and success could not be conducted entirely at the level of propositional contents.

There is an interaction between one's choice of semantic primitives (inference or representation) and one's choice of grammatical categorial primitives (sentences, or terms and predicates). The interaction is motivational rather than strictly conceptual, though—it is not that commitment to one semantic order of explanation entails commitment to a particular categorial order of explanation, or vice versa. Leibniz, who may serve as a paradigm for pre-Kantian inferentialists generally, begins his account with concepts standing in essentially inferential relations of inclusion to one another. Propositional contents are reached only by suitably combining these independently contentful items. So semantic inferentialism can coexist with a bottom-up categorial strategy.¹

Conversely, semantic representationalism is compatible with a top-down categorial strategy, which takes the fundamental sort of content to be propositional. Representing states of affairs, purporting to represent facts, need not be thought of as semantically decomposable. If the propositions represented are thought of, for instance, as sets of possible worlds, there would seem to be no necessity to continue by explaining the capacity to represent these things in terms of more primitive capacities to represent objects or properties.² Talk of objects and object-representings and properties and property-representings would then proceed in terms of role in propositions and proposition-representings (as it does for Kant).

The bottom-up categorial strategy is obliged to explain propositional se-

mantic contents at some point, however, for these are the contents expressed by sentences, the only expressions with which, as Wittgenstein says, one can make a move in the language-game. Failure to ascend to an account of such contents would disqualify a theory as a semantic theory, for it would sever that theory from any account of the use of linguistic expressions or the significance of beliefs. It is precisely the role it plays in explaining the proprieties of the use of linguistic expressions or the possession of intentional states such as belief that qualifies something associated with those expressions as a semantic content. Dually, the top-down categorial strategy is obliged to explain the subpropositional contents expressed by subsentential expressions such as singular terms and predicates. Failure to descend to an account of such contents would doom a theory to explanatory inadequacy, for it would then be able to make no sense of the connection between saying something (expressing a proposition) and talking about something (characterizing an object). While these two ought not to be identified at the outset, the latter phenomenon is too central to our understanding of what we are doing when we talk and think simply to be ignored. Unless it accounts for the possibility of representing particular objects, a semantic theory will not address the concerns that many have taken to define its topic. The relation between these categorial strategies may be compared in this regard to that between inferentialist and representationalist commitments to fundamental semantic concepts. Each reductive order of explanation must account for the notions treated as primitive by the other, or independent accounts must be offered of each sort of primitive, together with a theory that specifies how they collaborate.

2. Freestanding and Ingredient Contents

The conclusion is that any account of the representational character of propositionally contentful states, acts, and utterances is obliged to offer a reading of singular reference (the representation of particular objects) and of property-representation. For the link between belief and particular objects is a sort of paradigm of representational directedness. Offering such a reading requires looking at subpropositional contents and the way in which one expression can occur as a semantically significant component in another. The only sort of contentful expressions that have been officially discussed so far are sentences (and a very special sort of sentential operator). So it will be well to begin by considering the concept of sentential embedding in general: how the content of one sentence can contribute to the content of a compound sentence in which it is embedded as a semantically significant component. Starting with the special case in which the only subsentential components considered are themselves sentences has the advantage that this grammatical category can already be specified and understood in terms of another aspect of its use, its directly inferential significance in expressing assertional commitments. Thus a sense can be given to the question, What is the relation

between the sort of content relevant to this fundamental assertional and inferential use of sentential expressions and their derivative use as (in general) unasserted components of assertible sentential expressions? With a grip on this relation it will be possible to move on to consider the contents of expressions whose *only* use is as unasserted components of sentences, paradigmatically singular terms and predicates.

The primary job of a concept of semantic content, it has been emphasized, is to account for the pragmatic significance of the states, performances, and expressions that are understood as exhibiting such contents. The more specific theoretical commitments that have been forwarded so far are intended to fill in notions of content and significance that can satisfy this basic principle. These subordinate endorsements include the practical and normative understanding of those significances in terms of deontic states, the social-phenomenalist understanding of those deontic states in terms of discursive scorekeeping by adoption of socially perspectival deontic attitudes, the idea that the sort of practice or use to begin with is linguistic, the idea that linguistic practice is distinguished by its government of assertional performances, the idea that assertional uses are essentially inferentially articulated, and the idea that inferential involvements correspond to propositional contents.

Frege builds a basic structure of semantics and pragmatics into his system from the beginning, distinguishing accounts of the significance of judging, under the heading "theories of force," from accounts of the contents judged, under the heading "theories of content." As part of his specification of the task of the theory of content, Frege recognizes that expressions can be contentful not only in the sense that a certain force can be attached to their utterance but also in the sense that their occurrence expresses something about the content, in the first sense, of sentences in which they appear.³ As Dummett puts the distinction:

In speaking of sentences themselves there are two different ways in which we may regard them; and these may give rise to two distinct notions of [content]. On the one hand, we may think of sentences as complete utterances by means of which, when a specific kind of force is attached, a linguistic act may be effected: in this connection, we require that notion of [content] in terms of which the particular kind of force may be explained. On the other hand, sentences may also occur as constituent parts of other sentences, and, in this connection, may have a semantic role in helping to determine the [content] of the whole sentence: so here we shall be concerned with whatever notion of [content] is required to explain how the [content] of a complex sentence is determined from that of its components. There is no *a priori* reason why the two notions of [content] should coincide.⁴

It is this second notion, and its relation to the first, that is the current topic.

The technical terms Dummett introduces to capture the two dimensions

of sentential content that Frege discerned are "freestanding sense" and "ingredient sense." Each of these indicates an explanatory role that the notion of content as truth conditions has been thought to play: settling, in context, what the assertor of a freestanding (unembedded) sentence with that content thereby becomes committed to, and settling, in sentential context, the free-standing content of a compound sentence in which it is an (embedded) ingredient. Understanding these relations is particularly important from the point of view of a strategy, such as the present one, that seeks to work backward from notions of commitment and inference to notions such as truth conditions and representation. How should the notion of ingredient content be understood, and what does it have to do with talk about truth?

To begin with, it may be pointed out that in the passage above, where the bracketed word 'content' has been inserted, Dummett writes "truth-value." He is discussing Frege, and in the semantics of Frege's extensional logic, the concept of truth-value plays both sorts of role. Truth is what matters for the force of assertions of freestanding sentences. For it is what is preserved by good inferences, in particular the inferences that are good in virtue of their logical form—the ones Frege is codifying. Furthermore, possession, by a logically compound sentence, of the property preserved by logically good inferences is determined by the truth-values of its component sentences. When the same formal apparatus is maintained as much as possible—consistent with letting different notions play these two roles—the result is classical multivalued logic.

3. Multivalued Logic

The standard way of presenting these semantic ideas is as part of a bottom-up compositional definition of logical connectives, and of the validity of compound sentences formed by their use. The semantics is provided by a generalization of truth tables, defined not over Frege's two (truth) values but over many, perhaps an infinite number. Corresponding to each n -ary syntactic compounding device is a function mapping n -tuples of values assigned to component sentences onto the value assigned to the compound sentence in which they are components. These functions are most easily visualized in the form of the familiar sort of table:

!	[*1]	[2]	[3]
[*1]	1	2	3
[2]	1	3	3
[3]	2	3	1

According to this table, for instance, an interpretation that assigns p the value [2] and q the value [3] must assign $p!q$ the value [3].

Since the original role played by the notion of truth-value is being bifur-

cated, it is best not to beg questions by continuing to employ it for one or the other of these notions. The values in the set $\{[1], [2], [3]\}$ may be called *multivalues*. One or more of the multivalues is distinguished or *designated* (indicated by the '*' attached to the multivalue [1] in the table). A compound formula is valid in virtue of its form in case it is assigned a designated value no matter what multivalues are assigned to its component sentences.

Designatedness here indicates whatever the force-relevant notion is, for instance truth or, more generally, what is preserved by good inferences (which might in another context be some sort of commitment or entitlement). According to such a scheme, an interpretation assigns each sentence two sorts of value: as designated or not, and as having a certain multivalue.⁵ The designatedness value includes everything that matters for the pragmatic significance of the freestanding uses of the sentence (to which assertional force can be attached) as far as it is represented by this formal apparatus. Differences between sentences that are assigned the same designatedness value (in the example, designated or not designated) are significant at all only insofar as they affect the designatedness of compounds containing them. The two undesigned multivalues in the example differ in that substituting one for the other changes not only the multivalues but the designatedness of some compounds containing them.

The standard, bottom-up direction of explanation exploits this apparatus to move from an antecedent set of multivalues that can be associated with sentences, and from functions antecedently associated with compounding devices, via a notion of designatedness, to attributions of formal validity. The same apparatus, however, can be exploited in the service of the converse, top-down direction of explanation. Then the move is from antecedently understood attributions of material designatedness to assignments of multivalues to sentences and of functions to compounding devices. The essential principle is that if two sentences have the same multivalue, then substituting one for the other never changes the designatedness of any compound sentence in which they can appear as components. This is what is meant by saying that the multivalues express the contribution a sentence makes to the designatedness value of compounds containing it.

Since any sentence can be regarded as a degenerate compound containing itself, it follows from this principle that two sentences with the same multivalue must have the same designatedness value. This is what justifies the usual procedure—embodied both in the standard tabular way of setting out semantic definitions of connectives in multivalued logic and in the definitions that generalize it to semantic matrices—of treating multivalues, rather than sentences, as what take designatedness values. Turning the basic principle around, two sentences can be treated as having the same multivalue just in case substituting one for the other never changes the designatedness value of a compound sentence in which one appears as a component. In this way, sentences are assimilated into co-multivalue classes—and so taken as

sharing their ingredient contents—accordingly as their intersubstitution as components of compound sentences preserves the designatedness values of those compounds. Lindenbaum can be understood as employing an extreme form of this strategy in his mechanism for constructing, from the set of theorems of a logic meeting certain general conditions, a matrix of multivalues and compounding functions defined over them that would validate just those theorems, by identifying multivalues with equivalence classes of logically interderivable sentences, and designatedness with theoremhood. There is no guarantee that this procedure will not end up with an infinite number of small equivalence classes (as it does in the standard Lindenbaum algebra for the propositional calculus).

How finely the ingredient contents are individuated by this substitutional test depends on the expressive power of the language, specifically on what sentential embedding contexts and embedded sentences are discerned in it. Strictly speaking, substitutional assimilation according to multivalues or ingredient contents is always relativized to a class of embedded sentence occurrences, and so to a class of sentential embedding contexts. It is not implausible that in natural languages, for any two lexically distinct sentences, there is some context in which substitution of one for the other can affect the assertional designatedness of the compound sentence resulting from such substitution. For instance, 'S now thinks (or wishes) that *p*' is quite discriminating. This fact need not rob the substitutional form of analysis of its usefulness, for relative to various restricted classes of contexts, important assimilations are brought about nonetheless. Indeed, the partial ordering on sentential contexts that is brought about by looking at proper-inclusion relations among the multivalued equivalence classes they generate can contain interesting information about the semantic relations between those compounding devices.

Two embedding contexts can generate the same multivalues (in case they sort possible embedded sentences into just the same equivalence classes), or one can cut finer than another. Suppose, though, that every sentential embedding context that is discerned yields a different way of carving up the embedded sentences into equivalence classes, in a crazy-quilt of overlapping classes exhibiting no substantial identities or inclusions. In that case there would seem to be no theoretical advantage to discerning the semantically significant occurrence of one sentence in another. Occurrence of a sentential expression as a lexical part or syntactical subunit of another sentence is neither necessary nor sufficient to make it appropriate to discern the semantically significant occurrence of one sentence in another. The theorist may discern such occurrences where there are no lexical sentences, as with an embedded expression such as 'Kant's claim about *Aufklärung* and responsibility', or may deny them where there are, as Quine would do with direct quotation of sentences uttered. Discerning subsentential structure is enlightening only insofar as the assimilation of embedded sentences shows how the

capacity to use the embedded sentences, together with the capacity to use some of the embedding sentences, could generalize to a capacity to use the compound sentences with arbitrary embedded components.

It is worth considering a somewhat different, but closely related, point that Dummett makes, in connection with an approach to linguistic theorizing he associates with Wittgenstein:

One way in which these passages from Wittgenstein may be taken is as rejecting the whole idea that there is any one key idea in the theory of meaning: the meaning of each sentence is to be explained by a direct characterization of all the different features of its use; there is no uniform means of deriving all the other features from any one of them. Such an account would have no use for any distinction between sense and force: while it could admit some rough classification of sentences, or particular utterances of sentences, according to the kinds of linguistic act effected by means of them, it could cheerfully regard the totality of such types of linguistic act as unsurveyable—as Wittgenstein does—and would not need to invoke the classification of linguistic acts in its accounts of the meanings of particular sentences . . . The difficulty with such a theory is to see how it could do justice to the way in which the meanings of sentences are determined by the meanings of the words which compose them. The great strength of a theory which admits something as the key concept for the theory of meaning—at least a theory which is as developed as that of Frege—is that it displays a plausible pattern for the determination of the meaning of a sentence by the meanings of the constituent words . . . If nothing is to be taken to be a key concept, then we are once more without any conception of what the meaning of a word, as opposed to that of a sentence, is taken to be.⁶

The idea is that understanding a word need consist only in understanding the contribution it makes to the sense or content of sentences containing it. From there, the speech-act theory is to explain how that content contributes to the force or significance of various sorts of performances involving it. Otherwise, understanding the word requires mastery of the contribution it makes to all of the different acts that can be performed by means of it.

What underlies the analogy

	force : sense
	(or pragmatic significance : semantic content)
	::
	sentential content : subsentential content

is the thought that each variety of significance a performance can have sorts sentences that may be uttered with that force or significance into content-equivalence classes accordingly as intersubstitution preserves it. Unless dif-

ferent kinds of force or significance sort sentences into content-equivalence classes in the same way, no theoretical advance is made by discerning contents in addition to significances. Just so with assimilation of subsentential components preserving the content (or significance) of sentential compounds.

It was said above that although Frege is the first to take seriously the requirement that some aspect of semantic content determines the contribution a contentful expression makes to compounds in which it occurs as a component, nonetheless in the semantics for his logic he employs one notion, truth-value, to play both freestanding and ingredient roles. He there codifies inferences that depend only on the logical form of the sentences involved, and not on their material or nonlogical content. For these purposes he finds that it is possible to treat truth- (or commitment-) preservation not only as necessary for goodness of inference but also as sufficient. Thus for the inferences codified by his classical conditional, not only is designatedness preserved by good inferences, but any inference that preserves designatedness is a good one. For this compounding device, the two-valued conditional, sameness of designatedness value (which does duty here for freestanding content) is sufficient for sameness of multivalue (which does duty here for ingredient content). In this sentential context, the force-relevant content determines the role of sentences as components as well.

Designatedness-functional contexts such as this may be said to embed *homogeneously* with respect to designatedness values, since those values are all that matter in determining the contribution made by an embedded sentence to the designatedness value of the whole (that is, they can serve as multivalues). This term is used to mark off one of the several distinct senses sometimes attached to the expression *extensional*. Whether or not a sentential context is homogeneous in this sense concerns the relation between designatedness values and multivalues. It is quite independent of any specific conceptions of what plays the role of the immediately pragmatically relevant freestanding content. That role could be played by an antecedent concept of truth values or (looking ahead) by a concept of truth conditions.

Truth is preserved by good inferences of a certain important class. That class can be thought of as corresponding to deductive inferences, provided the notion is broadened beyond the concern with formally good inferences that is traditionally tied up with the notion of deduction. (Here the principle that a good inference never leads from premises that are true to a conclusion that is not true is being thought of as only a necessary condition on the goodness of inferences.) The inferences in question are just the commitment-preserving ones, ('committive inferences', for short). The pragmatic force of freestanding utterances of the expressions that can take truth-values of the sort preserved by good inferences (that is, sentential expressions) is assertional commitment, overtly acknowledged by, and so appropriately attributed to, the utterer. If the inference in question is a good one of this sort,

then to be committed to the premises is to be committed to the conclusion. No further understanding of the notion of truth-value in its role as designatedness value is required in order to proceed to the assimilation of sentences into multivalue equivalence classes.

Start with any set of concomitantly attributed or undertaken commitments to claims, some of which are expressed by sentences that are sententially compound (in that substitution for sentences that occur as their components makes sense). Then assign to two sentences the same componential value or multivalue (relative to that set of commitments and that compounding vocabulary) just in case substituting one for the other never turns a sentence expressing a claim in the set of concomitant commitments into a sentence expressing a claim that is not in that set. It follows that if two sentences are componentially equivalent, then they have the same designatedness value—the commitments in question include either both of them or neither.

To recap: The sort of content that has been considered here previously is the broadly inferential content that determines the correct uses of freestanding sentential utterances, paradigmatically the significance of asserting them. Content understood in this way can be associated only with expressions whose freestanding use has a pragmatic significance. It is not available as an interpretation of the contribution made by the occurrence of essentially subsentential expressions, such as singular terms and predicates. Following Frege and Dummett, a further sort of sentential content, 'ingredient' content, has been discerned, corresponding to the role that sentences can play as components of compound sentences. Although freestanding content may play the role of ingredient content (in homogeneous contexts), in general the latter is not reducible to the former. Ingredient contents are a sort that can coherently be attributed to expressions functioning only as components of assertible sentences, although so far only the contents to be associated with sentential sentence components have been considered. In the usual synthetic use of contents as multivalues, to define logical connectives, one begins with contents of this sort and determines designatedness values and, eventually, formal validity by their means. But the same apparatus can be exploited analytically, to move down from a notion of formal validity (as Lindenbaum does) to the assimilation of sentences according to their componential roles or, as has just been seen, from a notion of material designatedness (for example as assertional commitment) to multivalue equivalence classes. The mechanism whereby a simple notion of ingredient content (multivalue) is extracted from a simple notion of freestanding content (designatedness value) is purely substitutional. Two expressions are assimilated as making the same contribution to compound sentences in which they occur relative to some property of freestanding sentences just in case substituting one for the other never changes an embedding sentence from one that has the property to one that does not. As concern shifts to material, rather than formal, issues,

validity ceases to be the key notion, and designatedness comes to the fore, as the topmost property with respect to which substitutional invariances are assessed. The route to the notion of semantically significant occurrences of subsentential expressions, then, goes through the notion of substitution.

II. SUBSTITUTION, SENTENTIAL EMBEDDING, AND SEMANTIC ROLES

1. *Substitution and Subsential Content*

Frege's notion of *substitution* is the key to appreciating the characteristic theoretical role played by concepts of semantic content. This point begins to emerge when it is noticed that in the story just told, the relations between designatedness and multivaluedness, on the one hand, and between validity and designatedness, on the other, are of the same general sort. The fundamental pragmatic status that a notion of content is to help keep track of is that of assertional or doxastic commitment. As the previous chapter argued, this is a notion sufficiently intimately tied to that of truth claim that whatever sort of content ends up accounting for the pragmatic significance associated with that status for that reason has credentials as explicating one important dimension of truth talk.

Given the general understanding of the relation between material and formal proprieties of practice that has been urged earlier, the concept of the formal logical validity of claims should be treated as derived from that of material assertional commitment. The means of derivation are straightforwardly *substitutional*: A (logically) valid claim is one, first, that is designated (to which one does or ought to undertake or attribute commitment) and one, second, that cannot be turned into an undesignated claim by any substitutions restricted to a special class of vocabulary. In the case being considered, the nonlogical vocabulary consists just in the component sentences, from which the compound sentence is conceived as resulting upon the application of a logical sentential connective. If a sentential context is not valid, in that not all substitutions preserve designatedness, then it may be substitutionally homogeneous (designatedness-functional), provided that substitution within codesignatedness classes preserves designatedness. If not, then intersubstitution within multivaluedness classes, substitutionally heterogeneous with respect to designatedness, by definition will preserve designatedness. Valid claims are just those special sentential contexts with respect to which the multivalued substitutional equivalence class assimilates all sentences. Ordinary claims—which are not valid with respect to substitution for components generally, nor with respect to codesignated components—are valid with respect to substitution for component sentences by sentences sharing a multivaluedness. The same substitutional structure is responsible for moving up from material assertional commitments to assertional validity and moving down

from material assertional commitments to multivalues or ingredient contents.

If logically valid sentential contexts are just those that assimilate all sentences into one single multivalue equivalence class, what is their special interest? They are of interest because the way those valid contexts are compounded out of other, nonvalid ones has much to teach about those nonvalid contexts, which include the basic sentential connectives. Any metatheory that identifies a logic with the set of its theorems is committed to understanding the semantics of logical expressions only insofar as it is expressed by the capacity of those expressions to enter into assertorically valid combinations. As was pointed out in Chapter 2, Dummett correctly argues that this is an unduly restrictive view of the subject matter of logic.⁷ He would identify that subject matter by reference not to the theorems characterizing a logic but to its derivability relation. For classical Boolean logic these two notions are equivalent, but in general the theorems need not settle the derivability relation. He shows how the apparatus of multivalues can be applied in the definition of valid inferences, and not just in the definition of valid claims. He is concerned with defining validity from antecedent sets of multivalues, that is with the synthetic rather than the analytic use of this substitutional machinery. And since his topic is logical validity, he is concerned only with formal, and not with material, inferences. But the point he makes carries over to the analysis of material inferences and the derivation of a notion of material ingredient content, where its real significance becomes apparent.

A move from material assertional commitment as designatedness to material inferential commitment as designatedness corresponds to the move Dummett recommends from formal assertional validity to formal inferential validity as the notion with respect to which substitutional equivalence is assessed. The suggestion is to look at inferential commitments and correctnesses of inference instead of, or as well as, looking at assertional commitments and correctnesses of claims. A condition on the individuation of sentential contents as inferential roles can be generated from the notion of goodness of inferences by considering two sequential applications of the methodology of substitution that generates multivalues from the designatedness of compound sentences. Extending Frege's usage, two claims can be said to have the same inferential content just in case substitution of a token of the one type for a token of the other never turns a good inference into one that is not good, no matter whether the sentence appears as a premise or as part of the conclusion of the inference.

This principle does not depend on the existence in the language in question of sentential operators producing compound sentences in which other sentences are embedded. An inference here can be thought of as a pair of sets: of premise claims and of conclusion claims. Inferences can be treated as themselves a sort of compound in which sentences can appear as embedded

components, and such inferences can be classified as 'designated' (good) or not. If attention is restricted to inferences involving only freestanding occurrences of sentences, the equivalence classes of claims defined by preservation of goodness of inference on intersubstitution within the class may be called 'freestanding inferential contents'. Inferential contents so defined are generated just the way multivalues are, except that instead of looking at the designatedness of compounds such as conditionals as what must be preserved by substitution, one looks at the goodness of inferences. They are the products of the first application to the analysis of inferences of the substitutional methodology suggested by multivalues. The result is just what Frege defined as "*begriffliche Inhalt*" (conceptual content) at the beginning of the *Begriffsschrift*, as discussed in Chapter 2.

Component conceptual contents can then be defined by a second application of the substitutional methodology that introduces multivalues—this time to a case where what must be preserved on substitution is not the designatedness of compound sentences but their inferential contents, which were constructed substitutionally by the first application of the analogy with multivalues. The result is to put into play two concepts of broadly inferential content: contents as the inferential potentials of freestanding sentential utterance (including both their employment as premises and as conclusions of inferences), on the one hand, and contents as the contribution a sentence makes to the inferential content of compound sentences in which it appears as a component, on the other. This latter sort of content, which may be called the 'component content' of a sentence, arises from considering substitution within compounds, rather than within inferences. Putting the two definitions together, it follows that two sentences have the same component content if and only if substitution of one for the other as embedded components of any compound sentence never turns from good to not good an inference in which the compound sentence appears freestanding. Assimilating sentences accordingly as their intersubstitution in inferences preserves the material goodness of inferences yields freestanding content equivalence classes, and assimilating them accordingly as their intersubstitution in sentential compounds preserves freestanding content, yields component or ingredient content-equivalence classes. On the side of assertional commitments and proprieties, beginning with material designatedness of compound claims yields one level of further substitutional assimilation, namely multivalues. On the side of inferential commitments and proprieties, the substitutional machinery can be applied twice—once to yield a notion of freestanding inferential content, and once again to yield a notion of component inferential content. (It is irrelevant for this contrast that in either case the top-level material notion, whether assertional or inferential, can also be used substitutionally to define notions of formal validity.)

The first step in generating the inferential hierarchy of substitutional levels of content was made by noticing that Frege's substitutional definition

of equivalence of conceptual contents from material goodnesses of inference is analogous in structure to the definition of equivalence of multivalues from truth-as-designatedness, that is, material goodnesses of claims. The main adjustment required for this analogy is that inferences must be treated as a kind of context in which sentences can appear embedded, as premises and conclusions, and which as a whole is assessable according to its correctness. The second application of the substitutional machinery is more closely analogous to the assertional designatedness-multivalue paradigm in that only substitution within compound sentences in which other sentences occur as components is envisaged. It is less closely analogous to the paradigm than the first step in that what is preserved as the test of assimilation (corresponding to multivalue equivalence in the assertional paradigm) is in the inferential case not an on/off property, designated or not designated, but possession of a certain freestanding inferential role or value—of which there are many, perhaps infinitely many. It is by way of preparation for this point that the initial account above of the relation between designatedness and multivalues speaks of intersubstitution within multivalue classes as preserving designatedness *value*, even where this just means preserving designatedness.⁸ The substitutional conceptual machinery as such is indifferent as to whether what is preserved is membership in a single class (the designated ones), alike for all compound sentences whose components are being varied, or membership in whichever element of some partition the compound sentence whose components are being varied belongs to. That there is no technical difference does not mean, however, that there is no difference in the explanatory value of applying the technical machinery.

Dummett robustly acknowledges the requirement that a notion of semantic content qualifies as such only by its relevance to the pragmatic significance of acts, for which asserting serves as a prototype. He is concerned to argue that this requirement means that the substitutionally topmost level of interpretation, the level of designatedness, must be two-valued or on/off, since what must ultimately be settled is whether an assertion is or is not correct (assertible).⁹ As a general point, this seems dubious—perhaps a normative pragmatics need not be founded on the application of the dichotomy correct/incorrect to performances such as assertions. The discursive score-keeping account offered here is substantially more complex, as not only are commitment and entitlement distinguished, but track is kept of which are undertaken and which attributed—all articulating various ways in which a claim or an inference can count as correct. Again, thinking of pragmatic status as what must be preserved upon intersubstitution of sentences sharing a semantic content may be too narrow a formal paradigm—perhaps semantic contents can determine the correctness of material inferences without having for that reason to be conceived as preserved by good inferences.

Whether or not this on/off requirement could be shown to apply to semantic interpretation generally, there is a sense in which it is satisfied by the

inferential hierarchy that is here laid alongside the assertional one that Dummett considers. The topmost notion there is the goodness of inference, which can be thought of as a yes/no, correct/incorrect, two-valued affair: of designatedness rather than 'designatedness values'. This is because, though the purely substitutional machinery does not require it, the topmost notion in these hierarchies is a pragmatic one, as Dummett urges; furthermore, the sort of pragmatics being pursued here is one of deontic status and social attitude (commitment and entitlement, attributing and undertaking), and these are conceived as either characterizing an individual or not. The three-leveled inferential hierarchy shows, though, that the assessments of correctness that generate this two-valuedness at the top need not be directed in the first instance at sentences. When one starts with inferences, sentences are assimilated into many inferential role-equivalence classes, not simply into those that are designated as correct and those that are not.

2. *Two Concepts of Extensionality*

The assertional interpretive hierarchy—of designatedness as substantive assertional commitment and multivalues as equivalence classes of component sentences intersubstitutable saving the designatedness of compounds—gave rise to a natural notion of extensionality for sentential contexts as consisting in componential homogeneity. In this sense a context is extensional if the multivalued equivalence relation need cut no finer than the codesignatedness classes. How does the componential notion of extensionality apply to the inferential interpretive hierarchy? Since sentences are not inferences, the relation between designated inferences and the inferential roles of freestanding sentences cannot be homogeneous. It cannot strictly be that all one needs to know about freestanding sentences in order to assimilate them in such a way that intersubstitution within the resulting classes will preserve goodness of inference is whether or not they are good inferences, for they are not inferences at all. It will be necessary to look elsewhere for an analog of this sort of extensionality at the top level of the inferential hierarchy. At the lower level, when what is at issue is the relation between the inferential roles of sentences and their componential roles, however, a notion of extensionality as homogeneity does apply, since it is sentences in both cases that are assigned such roles.

A sentential context in which sentences can appear embedded as components is extensional in the sense of being componentially homogeneous just in case substitution of one claim for another with the same inferential role never alters the inferential role of the compound sentence containing them. It is a criterion of adequacy on semantically explicating vocabulary—which has been picked out here as deserving to be called specifically *logical* vocabulary in virtue of playing that expressive role—that it generate embedding contexts that are inferential-role-functional or homogeneous in this sense.

Thus the inferential role of a conditional claim is to be settled by the inferential roles of its antecedent and consequent. Not all vocabulary is like this. In some of its uses, for instance, the inferential role played by the important, pragmatically explicating expression '*S* claims that *p*' in some speaker's mouth depends not on the inferential role played by *p* for that speaker but on the role it is taken to play for *S*.¹⁰

Componential homogeneity is a concept that has application only within a substitutional hierarchy. There sentences are assimilated at the lower level, as associated with one sort of semantic interpretant, in case substituting one for another does not alter the assimilation at a higher level. That assimilation corresponds to association of another sort of semantic interpretant with sentences that depend in some way upon the sentences substituted for. A different sort of reducibility can be conceived in terms of the relations between the two-leveled hierarchy of assertional semantic interpretation and the three-leveled hierarchy of inferential semantic interpretation. A particularly strong bond between the assertional and the inferential orders would be forged if the designatedness of sentences determined the designatedness of inferences involving those sentences. The topmost assertional level of interpretation (committed/not committed) would then assimilate sentences into inferential-role equivalence classes, and assertional multivalues would coincide with inferential component contents. Commitment to the goodness of inferences would be preserved by substitutions for premises and conclusions, provided those substitutions preserve assertional commitment.¹¹

Notice that the concept of multivalue is not equivalent to that of component content unless never turning a designated claim into one that is not designated is sufficient for never turning a good inference into one that is not good. Since in any case preserving designatedness is a necessary condition of a good inference, sameness of component content will guarantee sameness of multivalue, but not in general vice versa. Multivalues capture the contribution that component sentences make to only the designatedness-functional inferences involving the compounds they are embedded in. Commitment to the goodness of an inference in this sense of goodness is what is expressed by the assertion of a classical two-valued truth functional, so-called material conditional. In connection with assessments of the formal correctness of certain kinds of logical inferences, treating preservation of assertional commitment as sufficient as well as necessary is not an entirely useless strategy, as Frege shows. But the principle it embodies is simply false if applied to genuinely material inferences, whose correctnesses constitute the possession of material content by the assertible sentences that appear as their premises and conclusions.

That inferential commitments should be determined (in this substitutional sense) by assertional commitments regarding their premises and conclusions—which is an interhierarchy rather than an intrahierarchy relation—is, however, another sense that has sometimes been associated

with the notion of extensionality. Dummett, in the chapter in which he discusses the distinction between truth-value as designatedness and truth-value as ingredient or multivalue, considers the tension in Frege between two conceptions of *Bedeutung*. In one sense this technical term is used just to mean something like 'semantic role', defined by substitutional assimilations. In another sense the relation between an expression and its interpreting *Bedeutung* is understood to be modeled on that between a name and its bearer. At this stage in the present exposition, names and bearers are not among the concepts it is officially permissible to pretend to understand. Sentences, and their pragmatic and semantic correlates—that is, assertional commitments and inferentially articulated propositional contents—are all that are onboard so far. But *Bedeutung* notoriously embraces not only particular objects as referred to by singular terms but also truth-values, taken by sentences. The analog at the categorial level of sentences, to the tension at the categorial level of terms between a substitutional notion of *Bedeutung* as semantic role and a representational notion of it modeled on the name/named relation, is the 'tension' between notions of content derived from the inferential hierarchy and those derived from the assertional.

For substitution within assertional codesignatedness classes to preserve inferential designatedness is only one way in which the assertional interpreters might determine the inferential ones. Another possibility is that two sentences might have the same freestanding inferential role in case they have the same multivalue or assertional component content. Whether or not this is so depends on the expressive resources of the language, on what sort of sentence-forming locutions it makes available. Where these resources include conditionals, since these codify inferential commitments as explicit assertional commitments, the assertional designatedness of conditionals will vary with substitution of antecedent for antecedent and consequent for consequent, unless the substituends share their freestanding inferential roles. Different kinds of conditionals may codify different classes of inferences, each of which defines a correlative substitutional notion of inferential role. Where the inferences corresponding to that role are expressible by conditionals in the language, assertional multivalues must cut as fine as freestanding inferential contents.

Of course, the assertional multivalues may partition the sentences into even smaller classes, as they will if the language permits compound sentential contexts interpretable as having the form '*S* believes that if . . . then *q*'. Whether or not the expressive resources of the language suffice to establish a general determination of freestanding inferential contents by assertional multivalues depends not only on what conditionals exist but on how they behave. If only finitary conjunction is available, for instance, assertional multivalues adequately represent freestanding inferential contents in general only if the language is compact. Again, if infinitary conjunction is available, it must be able to form 'enough' conjunctions, and so conditionals. In any

case, it is clear that the proper order of explanation runs from inferential role to assertional codifying locutions (such as conditionals), to a multivalued defined substitutionally with respect to assertions formed by the use of those locutions, not the other way around.

That is, one does not start with an intrasentential notion of assertional multivalued and then use that to define conditionals, and those to define inferences. Freestanding inferential contents must be defined as part of the same conceptual package as assertional designatedness. Inferential commitment must be considered along with assertional commitment. For it is its inferential role that determines what asserting a sentence commits and entitles one to, and what could commit or entitle one to it. Apart from such inferential involvements, an assertional commitment would be without content. The assertional hierarchy of interpretation should not be conceived as independent of and antecedent to the inferential one. It may be noticed, furthermore, that even if sameness of assertional multivalued ensured sameness of freestanding inferential content, it would not follow that it ensured sameness of inferential component content, unless sameness of freestanding inferential role were sufficient for sameness of inferential component content, that is, unless the inferential hierarchy were componentially homogeneous.

If for these reasons the material-inferential interpretive hierarchy should not be seen as derivative from the material-assertional interpretive hierarchy, what about the other way around? Within these substitutional hierarchies there is a definite sense to the claim that one sort of content 'determines' another. Assertional multivalueds determine assertional designatedness in the sense that two sentences cannot have the same assertional multivalued and different designatedness values. It is this sense in which freestanding inferential contents determine inferential designatedness, and inferential component contents determine the freestanding ones. One cannot in the same sense ask whether inferential designatedness determines assertional designatedness, since there is no one sort of thing that can take both values. One can ask whether it is possible for two interlocutors to undertake or have attributed to them just the same inferential commitments but different assertional ones. Apart from commitment to conditionals, it would seem that this possibility ought to be allowed. Two scientists or two politicians might agree entirely about what would be true if certain conditions obtained but nevertheless have quite different beliefs about what conditions do in fact obtain. Though it is true that in classical two-valued logic, fixing the truth-values of all the conditionals (corresponding to inferential commitments, of a sort) determines the truth-values of all of the atomic propositions, this property is an embarrassment and provides further good reason to deny that the classic horseshoe means "if . . . then . . ." ¹² Hypothetical commitments ought not to settle categorical commitments. Rather, inferential commitments determine assertional commitments only taken together with other assertional

commitments. They cannot do the job all on their own. When purely formal inferences are at issue, there is a purely formal sort of assertion (namely of theoremhood) for which such determination can be envisioned: given a formal derivability relation M , one can consider $\{A: \Phi MA\}$, that is, the set of claims derivable from the empty set of premises. Of course the same set can be defined with respect to a material-inferential relation, but the resulting set of claims is not the only one compatible with the inferential commitments that generate it.

3. *Compositionality and Decompositionality*

The primary lesson that should be drawn from this discussion is that there is an intimate relationship between the notion of *semantic content* and the concept of *substitution*. That concept is one of Frege's grand themes, exploited everywhere in the official definitions of logical and semantic concepts. He is methodologically quite self-conscious about the importance of his substitutional approach—it is the basis for his technical concept of function, which in his most metaphysical writings he takes as an explicit topic for philosophical inquiry. Frege's earliest semantic and logical work introduces the concept of conceptual content in terms of substitutional behavior with respect to a kind of pragmatic significance: two claims have the same content if substituting one for the other never turns a good inference into a bad one. Goodness of inference is a pragmatic matter—in Fregean terms, a matter of force; in this paradigmatic case, it is a matter of the force of reasons. In the terms being recommended here, it is a matter of normative force, of deontic status, and so of social practice and attitude. However the pragmatic end is conceived, the route from pragmatics to semantics is that of assimilating expressions according to invariance (of pragmatic significance of some sort) under substitution. This same substitutional path that leads from inference to sentential conceptual content leads as well from the possession of freestanding inferential content by compound sentences to the possession of component-inferential content by embedded ingredient sentences and, as will appear in the rest of this chapter, from sentential content to the content of subsentential expressions such as singular terms and predicates. The substitutional way of working out a top-down categorial explanatory strategy is already implicit in the substitutional form taken by Frege's inferentialist approach to propositional content.

This *decompositional* methodology is what lies behind what is often called Frege's 'principle of compositionality'. According to that principle, the semantic interpretant associated with a compound sentence such as a conditional should be a function of the semantic interpretants associated with its semantically significant components. In spite of the way it is usually interpreted or exploited, this principle by itself is neutral between bottom-up and top-down categorial explanatory strategies. As Frege's own substitutional

understanding of functions indicates, the principle operates as a constraint on what it is for one expression to count as a semantically significant component of another, regardless of whether the compound is conceived as built up in the first place by operations on antecedently specifiable components or, conversely, the components are conceived as substitutionally precipitated out of antecedently specifiable compounds.

Typically, discussions of the compositional constraint are framed, not in terms of the generic notion of a semantic interpretant, as above, but in terms of the specific notions of sense and reference that Frege introduced in 1891. Standard sketches of the explanatory roles characteristic of those two semantic conceptions center on the following leading ideas:

1. The referent of a sentence is its truth-value, what must be preserved by good inferences.
2. The sense of a sentence is the thought it expresses, what is grasped by someone who understands it.
3. The referents associated with compound expressions are functions of the referents of their components.
4. The senses associated with compound expressions are functions of the senses of their components.
5. The sense of a sentence, together with how things actually are, fixes its referent, that is, its truth-value.

Great care is required in specifying just what commitments one is attributing to Frege under headings such as these, but the point to be made here concerns only gross structure. Again, if subsentential expressions were currently at issue, further doctrines would need to be included, most notably that the referent of a proper name is the object that sentences it occurs in are about, or on which their truth depends.

At the crude level of description expressed by these five dicta, it may be noted that there is a natural mapping of the substitutionally generated assertional and inferential semantic hierarchies onto the Fregean scheme of sense and reference. Corresponding to (1), assertional designatedness plays the role of truth-value as what must be preserved by good inferences. Corresponding to (2), what one must grasp in order to understand a sentence is conceptual content, the *begriffliche Inhalt* of the *Begriffsschrift*. Two claims are defined as having the same conceptual content in case substituting one for the other never turns a good inference into a bad one. If substitutions within compound sentences such as conditionals, appearing as premises or conclusions, are included (as the subsequent practice of the *Begriffsschrift* in fact does), then the restricted compositional principle (4) will obtain as well, for the ingredient inferential contents of component sentences determine the ingredient inferential contents of the compounds they occur in quite generally. If the relevant substitutions are restricted to sentences playing freestanding roles as premise or conclusion, then what must be grasped is only freestand-

ing inferential content. In that case the restricted compositional principle (4) will obtain only for substitutionally homogeneous contexts of embedding, that is, where sharing an inferential role is sufficient for sharing a component inferential content. In any case, the corresponding restricted compositional principle on the side of reference, (3), will obtain only where assertional codesignatedness is sufficient for multivalue assimilation, that is, in assertionally homogeneous sentential contexts.

Frege is the first to investigate systematically the concept of the special sort of semantic value or content that an expression can have in virtue of its contribution to the semantic value or content of compound expressions in which it appears as a significant component. The latter notion arises in the context of a substitutional understanding of semantic contents in their relation to pragmatic significances generally. The distinction between freestanding and ingredient semantic contents is blurred, however, by the fact that both on the side of reference (which in one of its explanatory functions has been identified here with the hierarchy of assertional contents) and on the side of sense (which in one of its explanatory functions has been identified here with the hierarchy of inferential contents), Frege seeks to have *one* notion from each hierarchy play both sorts of role. (Attention is still restricted for the time being to sentential interpretants, so this means truth-values, on the side of reference, and thoughts, on the side of sense.) A less confining semantic metatheory answering to the same insights expressed in the compositional requirements (3) and (4) above is achieved if two different sorts of content—freestanding and ingredient—are distinguished, both within the (de)compositional hierarchy generated by the semantic interpretants immediately relevant to assertional force, and within that generated by the semantic interpretants immediately relevant to the inferential content that is asserted.

What about (5), the structural principle that sense determines reference? Does the inferential role (freestanding or ingredient) of a sentence determine its assertional designatedness value? In one sense, clearly not, insofar as inferential agreement among interlocutors need not entail assertional agreement. In another sense, though, it does. Since the inferential role of the sentence determines what an assertor is committed to by it, one interlocutor cannot licitly assign to two sentences the same inferential role and different assertional designatedness values. The most interesting sense of this question, however, is neither of these. It is rather whether inferential contents determine assertional commitments in a way analogous to that in which truth conditions are understood to determine truth-values in the familiar conceptions of sentential semantics that take their cue from the later Frege's willingness to specify the senses expressed by sentences in terms of their truth conditions. Characteristic of these conceptions is an understanding of truth conditions as supplying one sort of sentential content: meaning as intension, as defining a function, which given a world or set of facts, deter-

mines the truth-value of that sentence (relative to that world or those facts). Model theory and formal intensional semantics provide representationalist, bottom-up implementations of such an understanding.

The role being played by truth-values in this sort of story is that of assertional designatedness. In the rival scheme being developed here, that role is played by the notion of assertional commitment. The notion of free-standing inferential content, which is derived substitutionally from that of inferential commitment (or designatedness of inferences), is, like that of content as truth conditions, intended to specify the content of sentences in the sense of what it is that can be assertionally designated or true. The present question is whether and in what sense inferential contents can serve the function that truth conditions serve, of determining, together with the facts, truth-values-as-designatedness-values. Can the contents substitutionally extracted from the pragmatic status of inferential commitment be construed as related to those substitutionally extracted from the pragmatic status of assertional commitment as intension to extension?

There is a sense in which inferential contents, together with the facts, determine what is true (assertionally designated). Thus component contents may be seen as corresponding on the inferentially intensional side to multi-values on the inferentially extensional, that is assertional, side. Component contents, which determine inferential contents, can thus be thought of as expressing the contribution sentences make to the truth conditions-as-intensions of compounds containing them. Further consideration of the issue must be postponed until Chapter 8, because it cannot be pursued while continuing to suppress the additional level of analysis at which the deontic status of assertional commitment (designation) is resolved into social attitudes, which are explicitly relativized as to attributor and attributee.

When loose talk of deontic status is replaced by careful talk about deontic social attitudes, the essential clues will be seen to be that facts are just true claims, and that taking-true is just asserting. For each interlocutor, the inferential contents associated with anyone's sentences, together with the facts, determine which of those sentences express truths. The facts consulted in each case are the claims the attributing interlocutor takes to be true (that is, endorses) or acknowledges assertional commitment to. If inferential commitments, and so inferential contents, were uniform across a community, the determined truths and the determining facts would in every case coincide (though unless assertional commitments were also uniform, they would vary from attributor to attributor). But this is an extraordinary and degenerate sort of case, one that bears the same relation to the fundamental situation of communication involving the practice of truth assessment that a community in which assertional commitments are universally shared bears to the fundamental situation of communication involving the practice of assertion. The full story of how the inferential content attached to an expression by an interlocutor affects the proprieties of truth attributions by others, and espe-

cially the role played by (what are treated as) the facts, of how not only sentential but subsentential sense determines reference, is presented in Chapter 8 as the centerpiece of an account of the significance of representational idioms for semantic theories.

4. Summary

Dummett explicitly distinguishes the explanatory role played by freestanding and ingredient contents and recognizes Frege's accomplishment in conceiving of the latter. He also explains the relevance of multivalued logic to these two notions of semantic content by pointing out that the two notions of truth-value in play in such logics—what have been called here 'designatedness' and 'multivalence'—ought to be understood just as versions of freestanding and ingredient content, respectively. These insights of Frege and Dummett have been applied and extended here by conjoining them with three further theoretical orientations. First, the substitutional apparatus that induces the distinction between levels of content is applied *analytically*, or in a top-down categorial direction, rather than *synthetically*, or in the bottom-up categorial direction of explanation that has dominated logic and semantics since Frege. Second, where standard treatments focus exclusively on the pragmatic goodness of *asserting*, to generate a top-level notion of truth-value as assertional designatedness, attention has been drawn here as well to the pragmatic goodness of *inferring*, to assign inferential roles to sentences, on the basis of which ingredient contents can then be defined substitutionally. Dummett recommends a move like this in understanding multivalued logics, under the heading of shifting from concern with logical validity to concern with logical derivability, from formally good claims to formally good inferences.

Finally, and perhaps most important, the substitutional mechanism that relates designatedness to multivalues is applied to *contents* rather than to *forms*. It provides a general semantic structure answering to material commitments and endorsements and the proprieties they induce, not merely to formal ones. On the assertional side, assertional commitments generally are considered at the top level, not just formally validated theorems. On the inferential side, material-inferential commitments are considered at the top level, not just formally valid inferences. Furthermore, this shift from the formal to the material is extended down the substitutional hierarchy—not just from pragmatic significance to freestanding sentential semantic content, but from such content to the ingredient contents that matter for the behavior of sentences as components in compound sentences. So the sentential compounding devices that can be considered are extended from purely formal vocabulary such as the conditional to any materially contentful sentential context in which other sentences can appear embedded as components (embedded as components in the sense that they can be substituted for). The

account that emerges adds another piece to the puzzle concerning the relations between logic and semantics, a piece that dovetails with the semantically expressive view of the distinctive task of logical vocabulary.

The problem with which this discussion began is generated by the fact that concepts of *semantic content* must, to deserve that appellation, play a role in determining the *pragmatic significance* of producing an utterance or adopting a state that exhibits such content. However, direct assertional and inferential significance attaches only to sentences, and furthermore only to sentences appearing freestanding, that is, as asserted or as premise or conclusion of an inference. Introducing the notion of substitution provides a model of a sort of indirectly assertional or inferential significance that the subsentential occurrence of an expression can have. This sort of content can be associated with sentences occurring as significant components in other sentences, rather than freestanding in an assertional or inferential way. Once this sort of ingredient content has been introduced into one's semantic theory, however, it becomes available to be associated also with expressions that (unlike sentences) can occur *only* as parts of assertible sentences. So the notion of *substitution* and substitutional content—which have been used here to cash out the notion of the 'contribution' the occurrence of a sentential component makes to the freestanding content of compound sentences it appears in—makes available a route into the assignment of broadly assertional and inferential contents to expressions of *subsentential* grammatical categories, such as singular terms and predicates, to which the concept of freestanding content does not apply. The rest of this chapter is devoted to considering the sort of semantic interpretant that can be substitutionally associated with such expressions.

A bonus arising out of this line of thought is that, while remaining entirely at the level of sentences, and while eschewing any appeal to notions of reference or representation (what Dummett calls "the semantic model of the name/bearer relation"), it has been shown how to make sense of the notion of extensionality of an embedding context. Dummett is wrong to say: "If the notion of reference were introduced in the first place simply as that of the semantic role of expressions of different kinds, without an appeal to the name/bearer relation as prototype, then, at the outset, we should have no inclination to distinguish intensional from extensional contexts, or to treat the former separately; on the contrary, there would be a natural presumption in favour of a uniform semantic treatment for all contexts."¹³ In fact, two different (though related) senses have been specified in which a context may be called 'extensional'. One has to do with substitutional homogeneity, that is, the sufficiency of assimilations of sentences according to their freestanding role to serve as assimilations of sentences according to their ingredient role. The other has to do with the sufficiency of concepts of content extracted from the *assertional* substitutional hierarchy of freestanding and ingredient contents to do duty as concepts of content in the *inferential* substitutional

hierarchy of freestanding and ingredient contents. It is in this sense that an inference—thought of as a context in which sentences can appear as premises and conclusions—(and hence also the conditionals that makes that sort of inference propositionally explicit) can be called 'extensional'. The clues provided by these ways of conceiving extensionality will be exploited later in discussing the relations between *inferential* and *referential* semantics.

III. SUBSENTENTIAL EXPRESSIONS

1. *Singular Terms*

What conditions on the use of an expression are necessary and sufficient for it to be functioning as, or playing the role of, a singular term? What sort of expressive impoverishment is a language condemned to by not having anything playing that sort of role? The answers to these questions may seem straightforward, at least in the large. Singular terms are linguistic expressions that refer to, denote, or designate particular objects.¹⁴ The point of having something playing this role in linguistic practice is to make it possible to talk about particular objects, which, together with their properties and relations, make up the world in which the practice is conducted.

The first of these claims may be accepted without accepting the order of explanation presupposed by the transition from the first claim to the second. To begin with, it may be questioned whether the concept *particular object* can be made intelligible without appeal to the concept *singular term*. Frege, for instance, implicitly denies this when in the *Grundlagen* he explains the ontological category of particular objects, to which he is concerned to argue numbers belong, in effect as comprising whatever can be referred to by using singular terms, to which linguistic category he argues numerals belong. Again, it may sensibly be doubted whether the concept of singular reference is itself sufficiently clear to serve as an unexplained explainer. If it is not, then the responses offered above provide not so much answers to the questions they address as recipes for turning a suitable theory of reference into such answers. Insofar as one is sanguine about the prospects for such a theory, this of course is no bad thing. But it is important to be clear about what such a theory must account for in order to be serviceable in this explanatory context.

It is not enough, for instance, to explain only *successful* reference. For put somewhat more carefully, the first answer forwarded above must be that singular terms are expressions that, in Quine's useful phrase, "*purport* to refer to just one object."¹⁵ The qualification expressed in this slogan by the use of 'purport' has two different functions: to acknowledge the notorious possibility that a name or a definite description may *fail* in its referential bid, as 'the most rapidly converging sequence' (or 'the square root of 2', as opposed to 'the positive square root of 2') does, and to exclude *accidentally* singular